K SHREEKRISHNA UPADHYAYA

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PROFILE

AI/ML engineering student specializing in fine-tuning transformer architectures for speech audio classification. Experienced across the full lifecycle of applied deep learning, from optimized dataset pipeline preprocessing to agentic, specification-driven development workflows.

EDUCATION

VIVEKANANDA COLLEGE OF ENGINEERING AND TECHNOLOGY, PUTTUR DK, Karnataka
Bachelor of Engineering in Artificial Intelligence & Machine Learning September 2023 – Present
Cumulative GPA (till 6th semester): 8.69/10.0
Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures

PROJECTS

DADS (Detection and Analysis of Dysfluencies in Speech) | Python, PyTorch, Librosa September 2025 – November 2025
Stuttering dysfluency detection system on the SEP-28k dataset, exploring both per-type and multi-label CNN architectures.

- Trained 5 binary CNN classifiers (one per dysfluency type: prolongation, block, sound repetition, word repetition, interjection) on mel-spectrogram features with per-type hyperparameter search across window size, hop length, and mel bands
- Explored a CNN-LSTM multi-label classifier as an alternative to separate per-type models
- Built a PyQt5 desktop application with real-time audio recording, playback, spectrogram visualization, and background stutter detection inference

The Major Project extends this work with a Wav2Vec2 + CNN-transformer hybrid architecture.

FinDiary | Go, PostgreSQL, Flutter, gRPC July 2026 – Present
Ongoing personal finance tracking application developed entirely through Specification-Driven Development — specifications are authored and refined before an agentic AI workflow generates implementation code, with all output reviewed for architectural consistency.

- Architecting Go backend services with PostgreSQL persistence and gRPC-based client-server communication
- Directing development of a Flutter cross-platform client, with offline-first local state management planned
- Practicing spec-first, AI-assisted implementation as a deliberate alternative to traditional hand-written development, alongside the fully hand-written systems work below

Sn Project | C, x86_64 Assembly December 2025 – Present
Collection of 13 cross-platform C libraries (memory allocators, threading, logging, tracing) extracted from an experimental game engine into reusable, independently-versioned infrastructure.

- Designed 7 memory allocator strategies behind a unified interface; hand-wrote x86_64 atomics with explicit memory-ordering fences for cross-platform threading primitives

Snuk | C April 2026 – Present
Embeddable scripting language with a custom lexer, recursive-descent parser, and tree-walking interpreter (6K lines of C), including reference-counted memory management.

hash shell | C January 2026 – March 2026
Open-source contributor with 17 merged pull requests, including a self-proposed multi-PR refactoring initiative to align the codebase with upstream coding standards.

Additional Projects:
Regex Engine (Thompson NFA construction in C), StateFlow (DFA/NFA simulator with automaton validity checking), LC3VM (LC-3 virtual machine).

SKILLS

AI/ML: PyTorch, Transformers, Wav2Vec2, CNN Architectures, Model Fine-Tuning, Dataset Preprocessing & Evaluation, Specification-Driven / Agentic AI Workflows
Languages & Build: Python, C, C++, Go, Java, CMake, x86_64 Assembly, POSIX Shell
Development: Git, Linux, CI/CD (GitHub Actions), gRPC, REST APIs, PostgreSQL, Flutter, Typst
Achievements: Winner of Nexathon 2025 (SDIT) and DevHack 2025 (Sahyadri College)